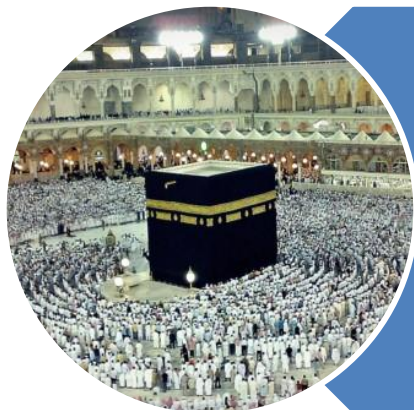


رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي



O my Lord! Expand for me my chest [with assurance] and ease for me my task and untie the knot from my tongue that they may understand my speech. **(Quran 20 : 25-28)**

Introduction to Data Structure

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What is Data?

- Data
 - A collection of facts from which a **conclusion** may be drawn
 - Example:
 - Temperature is 35 degrees celcius
 - Conclusion: It is HOT!
- Types of data:
 - Textual: for example
 - » Your name (Muhammad)
 - Numeric: for example
 - » Your id (090254)
 - Audio: for example,
 - » Your voice

What are Data Type?

- Data Type:
 - Built-in construct of any programming language to manipulate “**basic types of data**”
 - Basic Types of Data
 - Character
 - Integer
 - Real or Floating point number
 - Example (**Data Type**):
 - `int, char, double etc ;`

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What are Data Structures?

- Data Structure:
 - A particular way of storing and organizing data in a computer so that it can be used efficiently
 - It is a group of data elements stored together under one name
 - It is also referred as “Abstract Data Type (ADT)”
 - Example:
 - An array of integers
`int examGrades[30];`

Importance of Data Structures

- Goal:
 - We need to organize data
- For what purpose?
 - To facilitate efficient
 - storage of data
 - retrieval of data
 - manipulation of data
- Design Issue:
 - The challenge is to select the most appropriate data structure for the problem
 - Such is one of the motivations for this course

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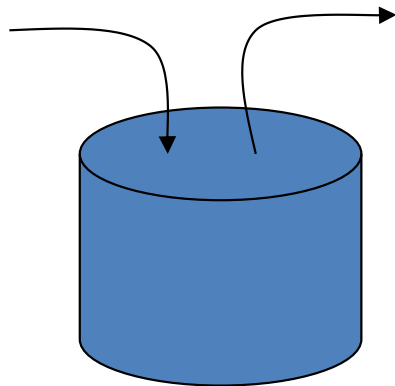
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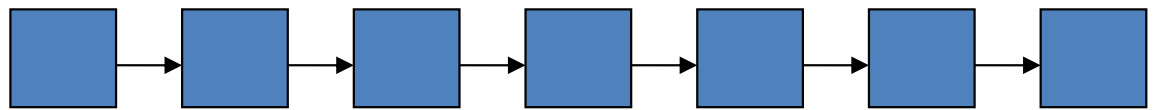
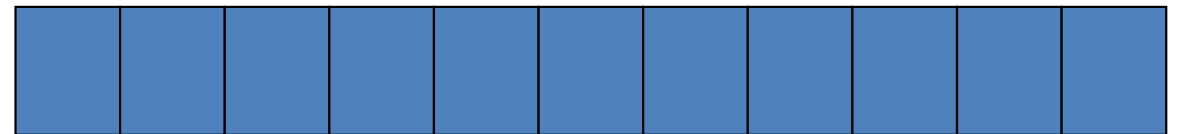
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List of Data Structures

- Arrays
- Linked List
- Stacks



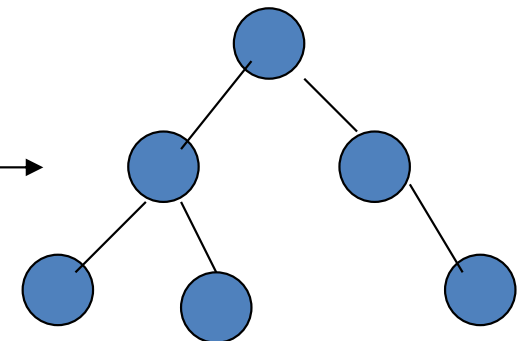
- Graph



- Queues



- Trees



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Classification of Data Structures

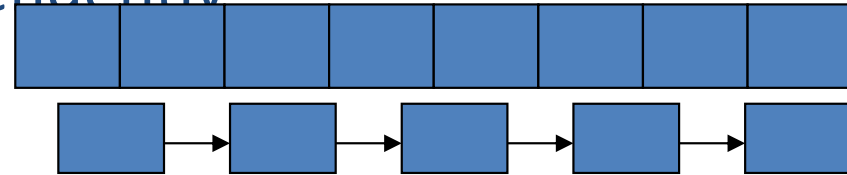
- Based on Existence:

- Physical data structures

- Can be created independently

- Array

- Linked List

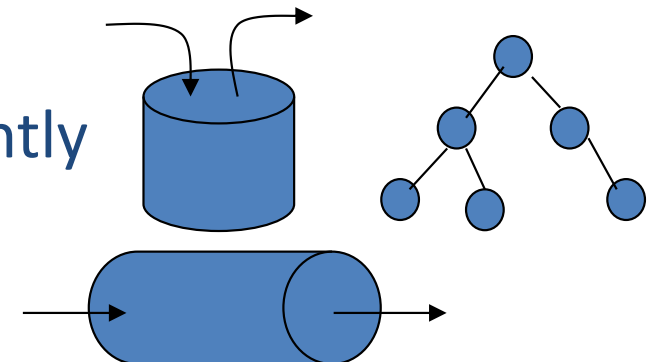


- Logical data structures

- Can't be created independently

- All others e.g.

- stack, queues, trees, graphs



Classification of Data Structures

- Based on Memory Allocation:

- Static** (or fixed sized) data structures

- Such as arrays



- Dynamic** data structures (change size as needed)

- Such as Linked Lists

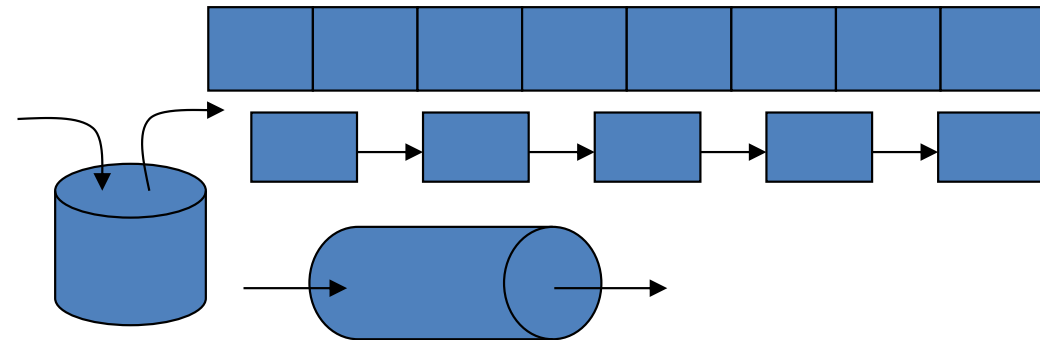


Classification of Data Structures

- Based on Representation

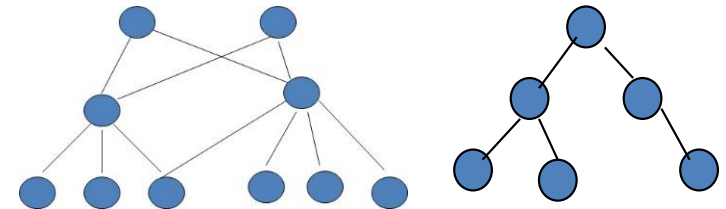
- Linear data structures

- Arrays
 - Linked lists
 - Stack
 - Queues



- Non-linear data structures

- Trees
 - Graphs



Operations on Data Structures

- Traversing
 - Accessing/visiting each data element exactly once so that certain items in the data may be processed
- Searching
 - Finding the location of a given data element (key) in the structure
- Insertion:
 - Adding a new data element to the structure

```
int[] arr = {10, 20, 30};
for (int i = 0; i < arr.length; i++) {
    System.out.println(arr[i]);
}
```

```
int target = 20;
for (int i = 0; i < arr.length; i++) {
    if (arr[i] == target) {
        System.out.println("Found at index " + i);
    }
}
```

```
ArrayList<Integer> list = new ArrayList<>();
list.add(5);
list.add(10);
list.add(20);
System.out.println(list);
```

Operations on Data Structures

- Deletion

- Removing a data element from the structure

```
ArrayList<Integer> list = new ArrayList<>();
list.add(10);
list.add(20);
list.add(30);

list.remove(1);
System.out.println(list);
```

- Sorting

- Arrange the data elements in some logical fashion

```
int[] arr = {5, 2, 9, 1};
Arrays.sort(arr);
System.out.println(Arrays.toString(arr));
```

- ascending or descending

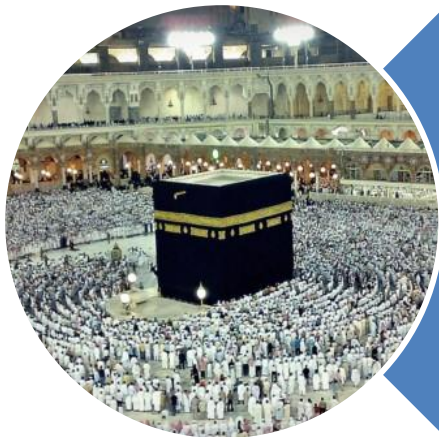
- Merging

- Combining data elements form two or more data structures into one

```
int[] arr1 = {1, 3, 5};
int[] arr2 = {2, 4, 6};
int[] merged = new int[arr1.length + arr2.length];

System.arraycopy(arr1, 0, merged, 0, arr1.length);
System.arraycopy(arr2, 0, merged, arr1.length, arr2.length);

System.out.println(Arrays.toString(merged));
```



Allah (Alone) is Sufficient for us,
and He is the Best Disposer of
affairs (for us). **(Quran 3 : 173)**